

KAIROS: AI-based Network Attack Forecasting (SIH PS 26153)

One line: KAIROS learns how network traffic EVOLVES -- $P(S_{t+1} | S_t)$ -- rolls it K steps forward, and warns BEFORE compromise with MITRE stage + evidence. Fully offline, explainable, reproducible.

Verdict on alignment: YES, point-by-point (table below). Honest gaps are stated, not hidden: 3/6 stages in selected CIC days; in-dist F1 trails baseline but wins the PS metric that matters -- 120s lead time; CTU-13 frozen scoring schema-blocked by design (no fake F1).

1. PS alignment checklist

- State as graphs/vectors -> per-10s host-flow graphs (kairos.graph.v1); CIC vector `__network__` + real CTU-13 topology.
- Learn $P(S_{t+1}|S_t)$ via GNN/Transformer -> GraphSAGE + causal Transformer, teacher-forced MSE next-state loss.
- Flow features (NetFlow/IPFIX) -> 80 CIC cols: flags, bytes/pkts, duration, IAT, ratios (IngestionService).
- Packet features (PCAP) -> TTL, window trend, frag, retrans, payload stats, seq/rand port-scan (C++ engine, 17.4M pkts).
- Train on labelled open data -> 800K capped flows -> 13,202 windows; labels from published CIC timelines.
- K-step rollout -> probability timeline $\text{Now}+T+1..T+K$ via dynamics.rollout + sigmoid head.
- MITRE stages -> 6-class softmax (5 PS stages + IMPACT) + heuristic cross-check.
- Explainability -> causal attention + TreeSHAP top-5 (R^2 0.9746), <170ms, never causal claims.
- Open + reproducible -> weights committed, seed 42, exact yaml/history/load test.
- Baseline comparison -> frozen LR vs world, identical split + strict + cross-day (benchmark_table.json).
- Offline demo (CSV/PCAP in, timeline out, no cloud) -> Java /forecast/upload + React console; Gemini optional-off.

2. Why static classifiers fail

STATIC: FlowA->label? FlowB->label? FlowC->label? (order discarded)
 Slow scan (1 port/10s) vs benign burst look identical in isolation.
 Reacts only AFTER malicious flow crossed the wire.

WORLD MODEL: $S_t \rightarrow P(S_{t+1}|S_t) \rightarrow S_{t+1} \rightarrow \dots \rightarrow S_{t+K}$
 current graph + causal history -> converges to compromise? -> WARN EARLY
 Infiltration is a PROCESS (probe->exploit->fanout->beacon->exfil), not one packet.

3. Core equation + architecture

Learn: $P(S_{t+1} | S_t)$. Roll: $S_t \rightarrow M(S_t) \rightarrow S_{t+1} \rightarrow M \rightarrow S_{t+2} \dots S_{t+K}$
 $M = \text{GNN (space: who talks to whom) + Transformer (time: what leads to what)}$.

PCAP/CSV -> [C++ packet feats --JNI-->] -> [Java 10s windows -> graphs v1]
 -> POST /predict -> [Python GNN->Transformer->K-rollout->sigmoid+softmax+SHAP]
 -> [Java narrative: offline-local DEFAULT; gemini-online only if ONLINE_MODE=key]
 -> [React: timeline + flagged table + stage overlay + badge Local/Gemini]

Browser never calls Python. Python never touches raw traffic. Gemini never touches prediction (narrative only).

4. What is a state?

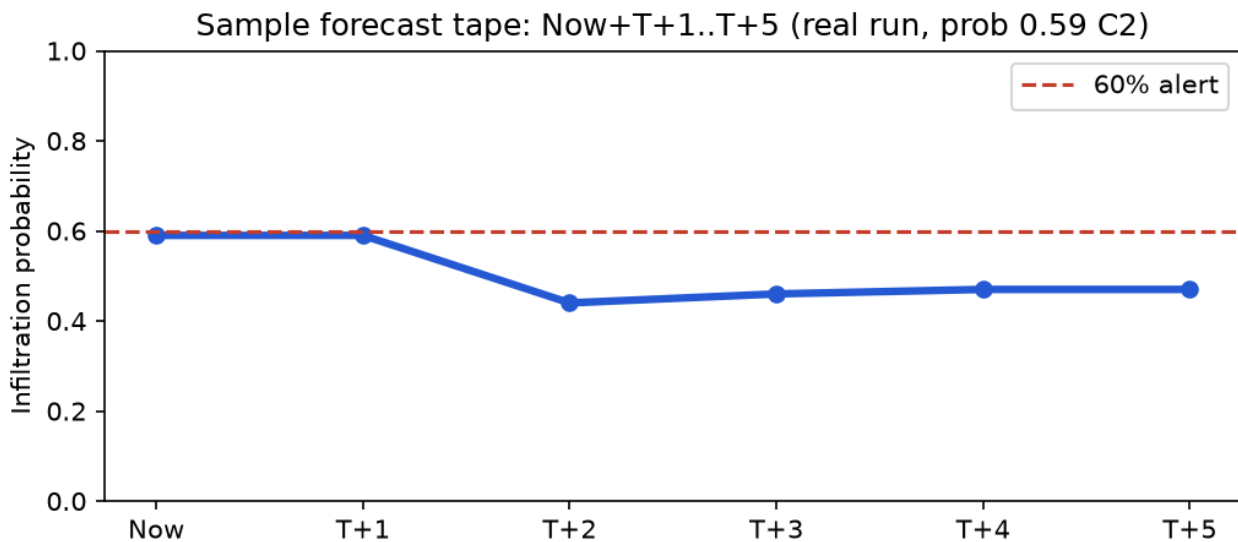
CIC vector (CSVs lack IPs, honest fallback):
`__network__ {flow_count, bytes, ports, syn/ack, topo=0}] topo=false`

CTU-13 packet (headers retained, real topology):
`[147.32.84.165] -> [10.0.0.5] {packet.*: count, ttl, window, payload...}`
 bounded 20k pkts -> 1263 flows, 443 hosts, 2 scan sources. topo=true

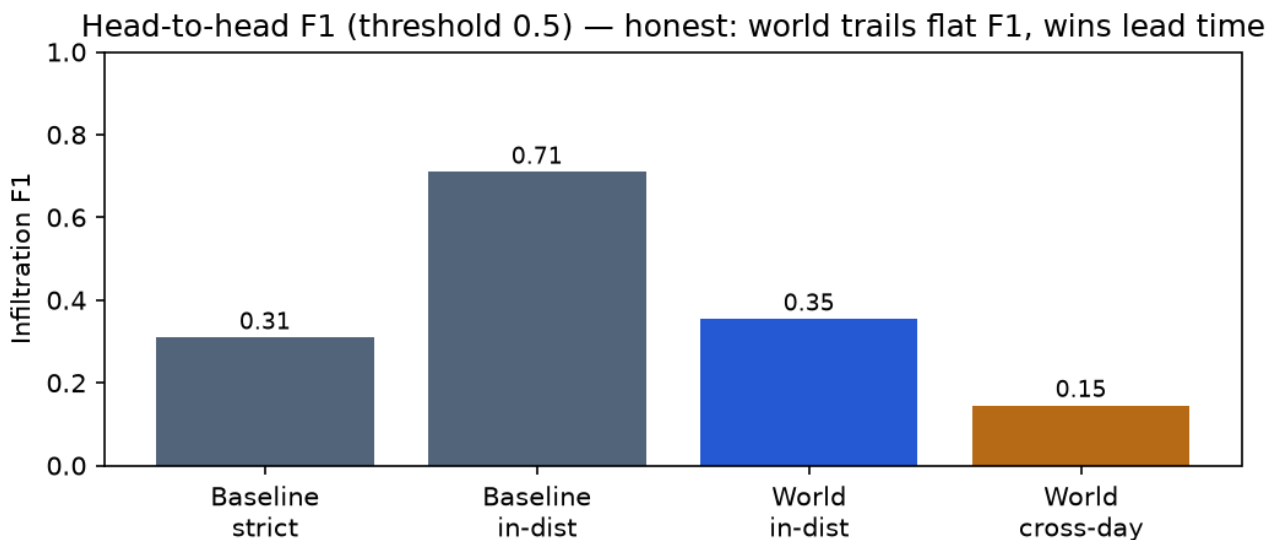
Same 8 node feats; edge schema differs -> loader REJECTS mismatch (no mis-score).

5. Live walkthrough (real numbers)

Input sample-attack.csv (10 rows benign->Infiltration) -> Java exports 10 windows -> Python K=5: immediate $p=0.59$ stage COMMAND_AND_CONTROL; rollout [0.59,0.44,0.46,0.47,0.47] stages [C2,IA,IA,IA,IA] max 0.59; attention 10 ctx future_mass 0.0 (causal); SHAP top drivers; latency ~200ms. Java local narrative: 'High risk 59.0% ...'. React: stamp 59.0% Below threshold (60%), blue Now->T+5, stage chips, evidence table up/down, green 'Local offline' badge.



6. Benchmarks (honest)



In-dist identical split: Baseline F1 0.7097 vs World 0.3545 -- world trails flat F1 (stated). Strict future holdout: Baseline 0.3096, stage 0.0 (481 C2 unseen -- honest failure). Cross-day: World 0.1455 (limitation kept). Lead time: World 120s (12 windows) vs Baseline N/A -- the PS proactive metric. CTU-13: schema-blocked, no F1 fabricated; Scenario6 excluded from scoring per manifest.

Takeaway for judges: the only system with early warning is the world model. F1 alone rewards memorization; lead time + explainability + offline is the PS ask.

7. Demo (offline, 3 terminals)

```
1) PYTHONPATH=python-ml .venv/bin/python python-ml/app.py          # :5000
2) (cd java-engine && ./mvnw spring-boot:run)                      # :8080
3) (cd react-ui && npm start)   # :3000 (no `npm run dev` in CRA)
Upload react-ui/public/sample-attack.csv -> timeline + evidence.
Proof: ONLINE_MODE=false ./scripts/verify_offline.sh (C++1/1 Java30 Py23 pass)
```